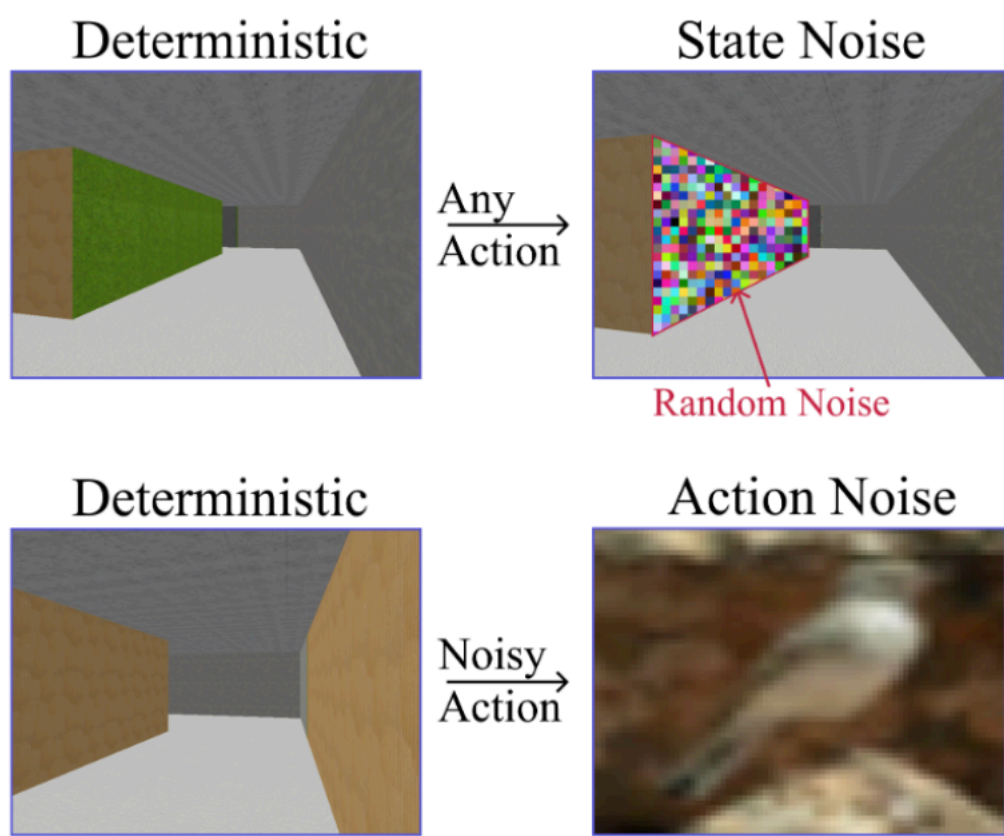


Exploration v.s. Exploitation

Intrinsic Reward Methods

Environments and Methods

- Envs: Clean, noisy-TV, action-noise



- Methods

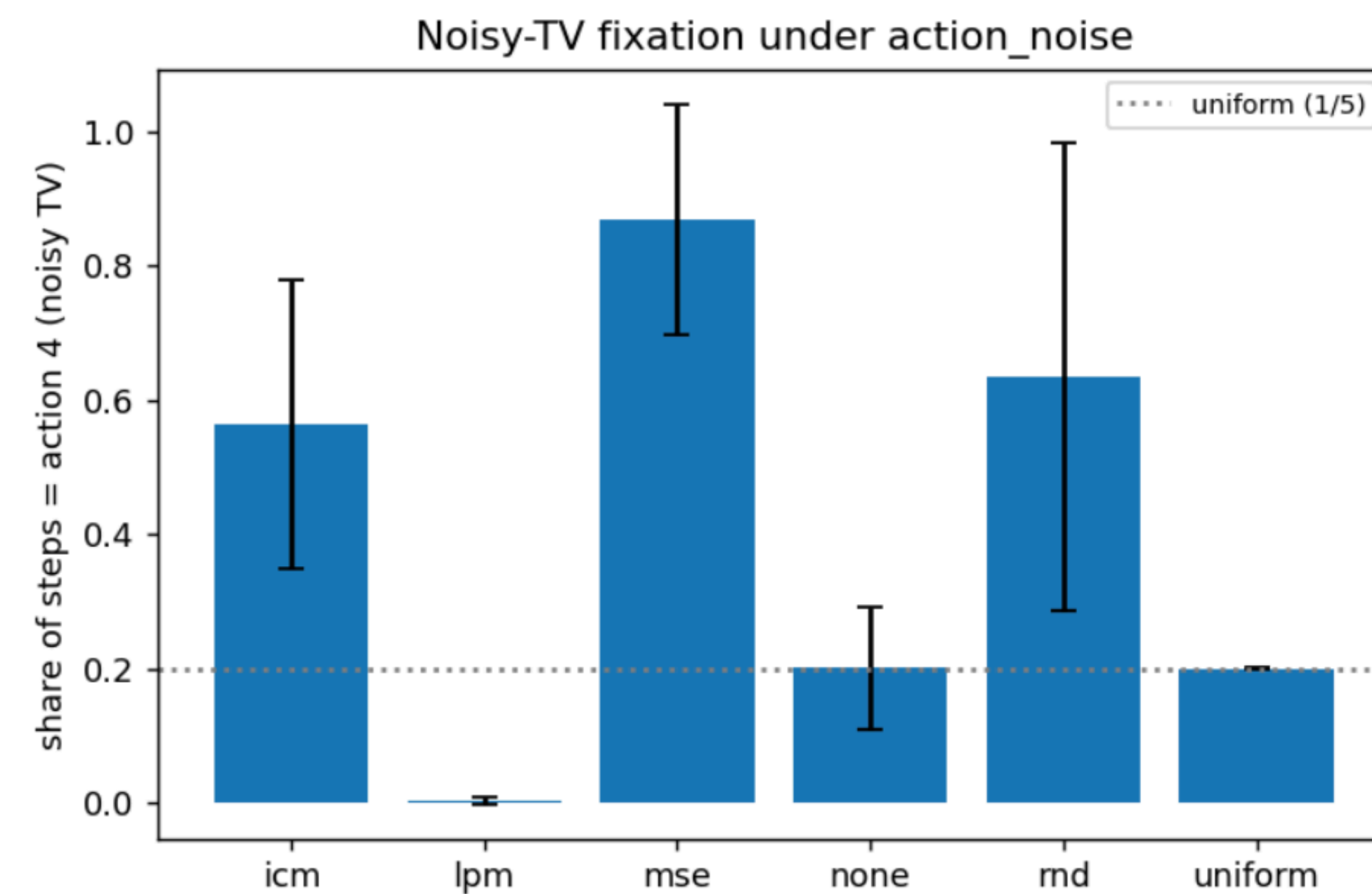
Class	Method	Mechanism (intrinsic reward r^i)
Learning progress	LPM	$\text{clip}(\eta \mathbb{E}[\varepsilon] - \varepsilon_{\text{actual}}, \leq 0.5)$: predicted-minus-actual next-state error (improvement)
Prediction error	RND	error of a predictor vs. a fixed random target network on the next frame
Prediction error (feat.)	ICM	forward-dynamics error in an inverse-dynamics feature space
Prediction error (pixel)	MSE	raw next-state pixel-prediction error (the pure noisy-TV victim)
None (baseline)	NONE	no intrinsic reward; A2C <i>is</i> still trained (entropy + value bootstrap)
Control	uniform	$a \sim \text{Uniform}\{0 \dots 4\}$; <i>no learning</i> (pure random walk)

- Grid experiments on 3 envs * 6 methods * 10 seeds

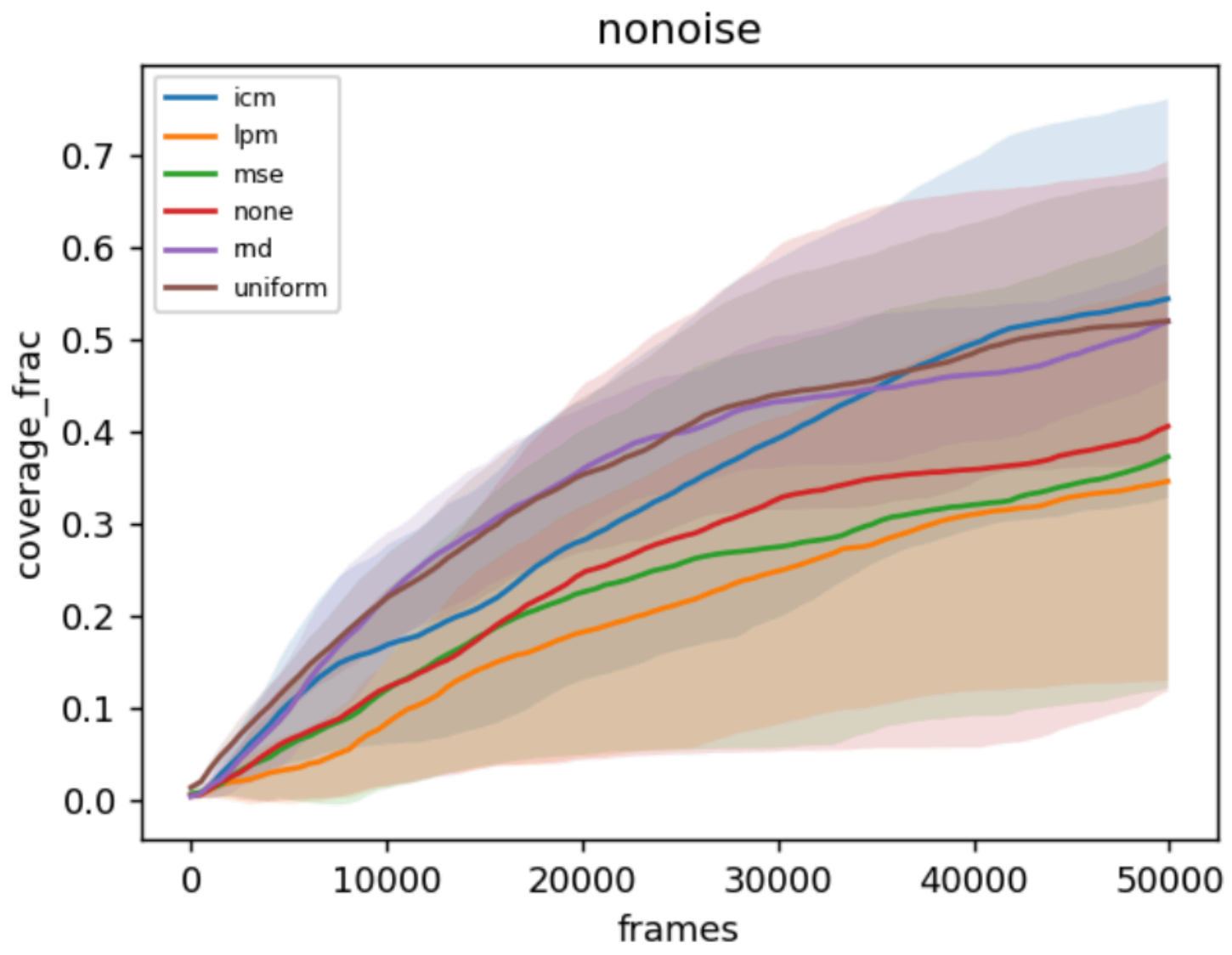
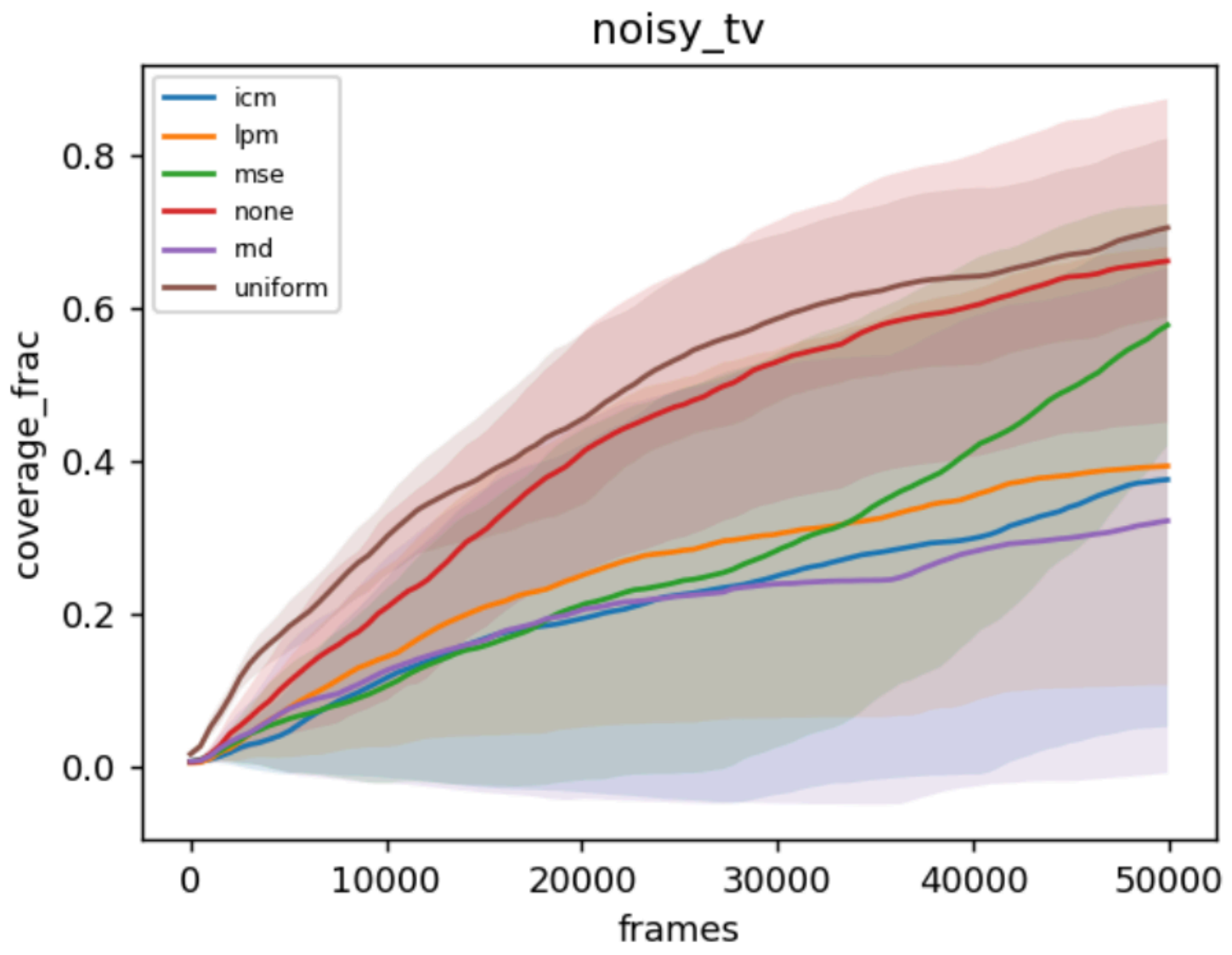
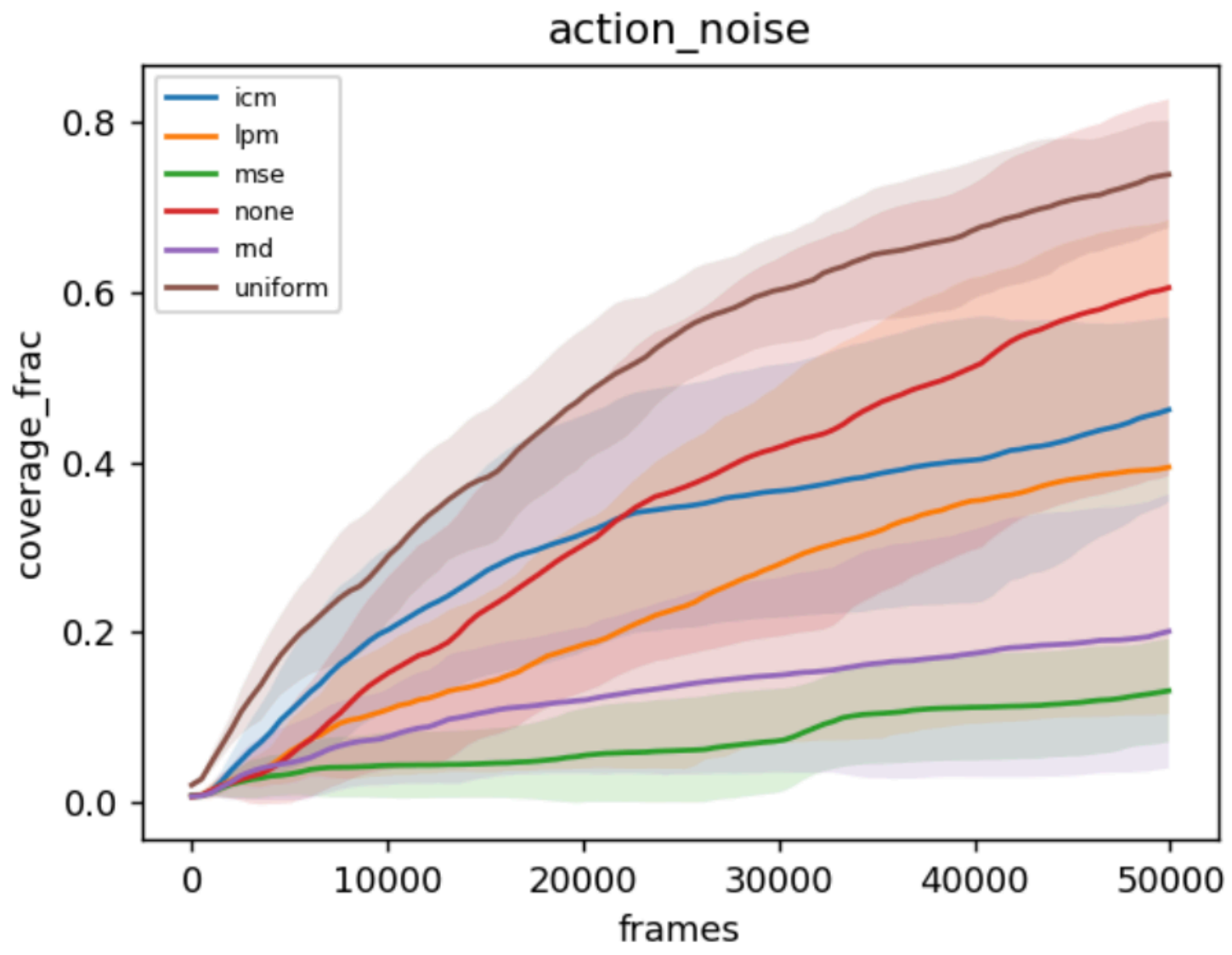
Metrics

- Coverage
- TV-action share
- Coverage-heatmap evolution

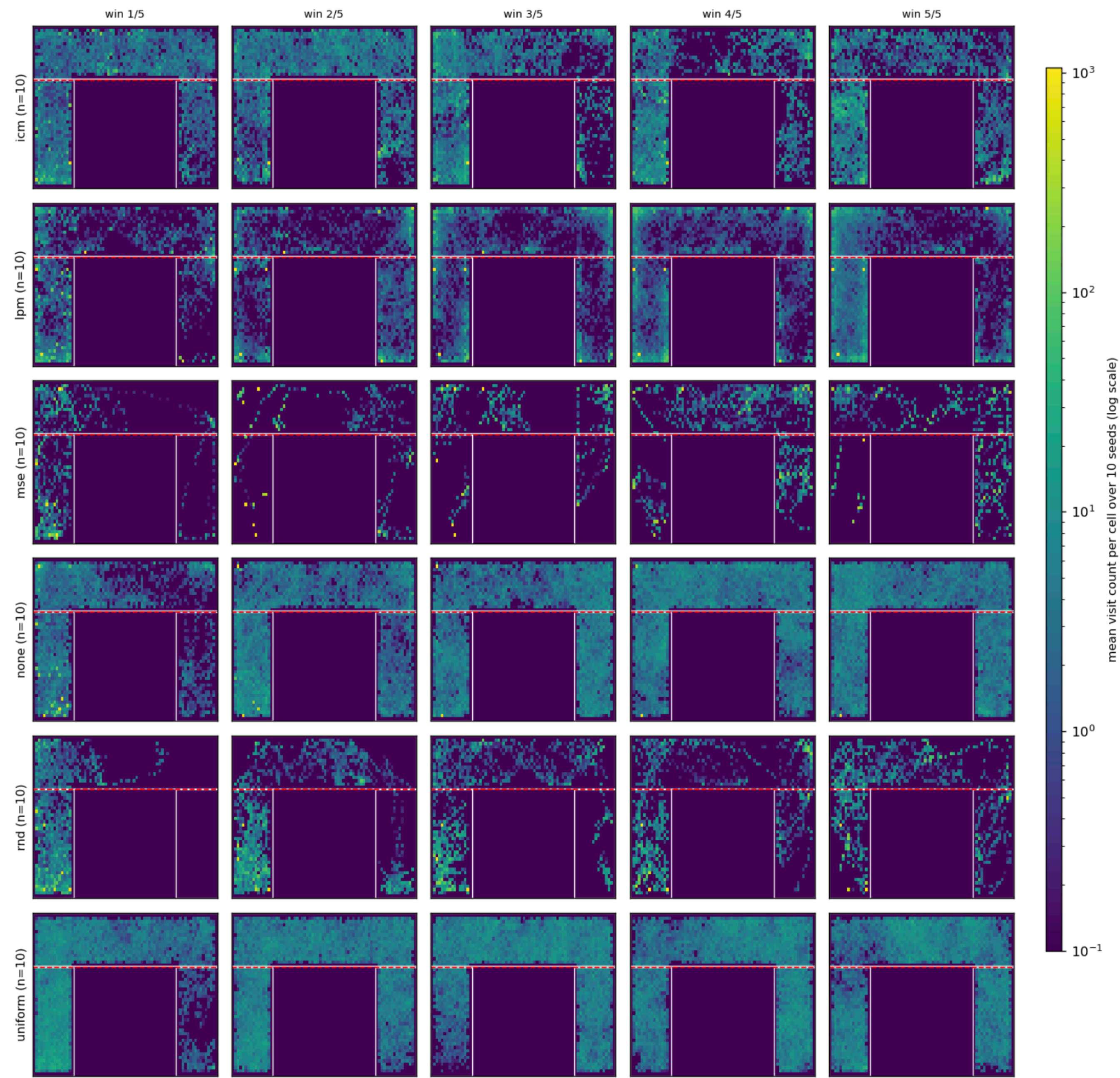
Policy	TV-action share (action_noise)
LPM	0.004 ± 0.005
uniform	0.200 ± 0.002
NONE	0.201 ± 0.091
ICM	0.563 ± 0.215
RND	0.634 ± 0.347
MSE	0.869 ± 0.172



Policy	nonoise	noisy_tv	action_noise
uniform	0.515 (0.16)	0.688 (0.12)	0.725 (0.07)
NONE	0.389 (0.31)	0.652 (0.22)	0.590 (0.23)
LPM	0.337 (0.22)	0.389 (0.30)	0.388 (0.30)
MSE	0.356 (0.26)	0.539 (0.20)	0.123 (0.06)
ICM	0.534 (0.23)	0.363 (0.34)	0.446 (0.13)
RND	0.501 (0.07)	0.311 (0.34)	0.192 (0.17)



action_noise — occupancy density (visit count), mean over 10 seeds



Next step

- Intrinsic reward methods on minigrid environment
- Why LPM is performing better on Ms PacMan. Verify the mechanism.

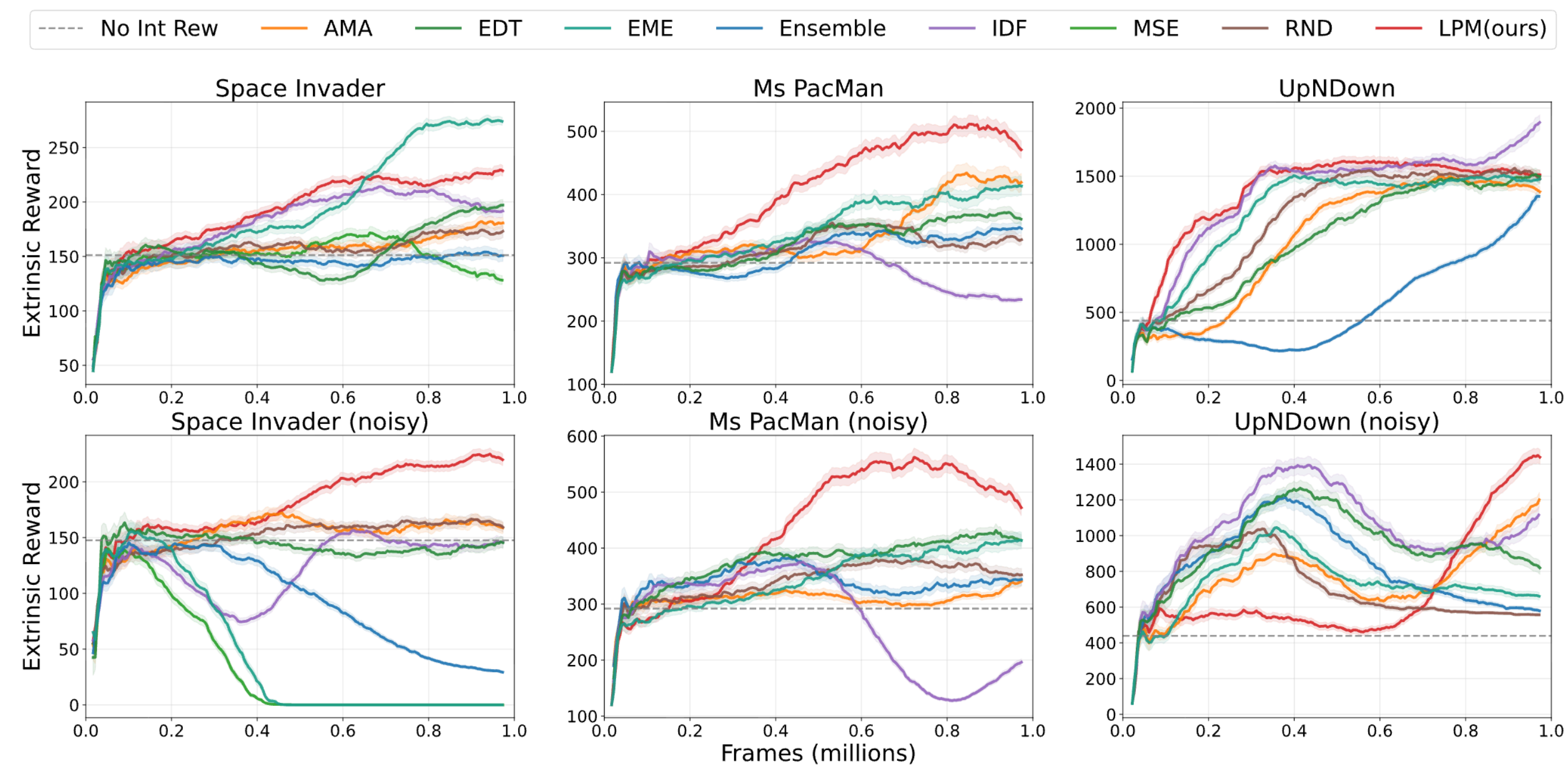


Figure 4: Average and standard deviation extrinsic rewards achieved by each method across deterministic and stochastic Atari environments with 128 random seeds.